VedaVenture

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Abstract

The traditional methods of learning arithmetic operations emphasize procedural sophistication and drilling. This leads to a lack of motivation and retention of students. However, with mobile learning and gamification becoming very popular for improving learning motivation by making learning accessible anytime and anywhere, there is still a lack of association with highly efficient computing frameworks. To address this issue, a complete mobile learning app was developed using Android Studio. This app has the potential to transform the mental training of students in math by educating and training them on Vedic Mathematics. Vedic Mathematics uses simple rules and patterns to simplify the process of computing. This makes it highly efficient for fast mental computation. This app integrates these highly efficient methods of mathematics with gamification and mobile learning space. A secure login system for protecting user data and elements of gamification and mlearning space are incorporated. These elements include points, learning levels, rewards, and learning streaks. Finally, with gamification and mLearning space incorporated with highly efficient computing methods, this app offers an incredibly interactive platform for improving learning and computation of students.

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1 Introduction

Mathematics plays a fundamental role in education, forming the basis for problem-solving, logical reasoning, and analytical thinking. However, many learners struggle with traditional arithmetic methods due to their procedural complexity and lack of engagement. Conventional teaching approaches often rely on repetitive practice and memorization, which can lead to reduced motivation and poor retention among students.

In recent years, digital learning technologies have emerged as effective tools to enhance educational experiences. Among these, gamification has gained significant attention for its ability to improve user engagement and motivation by incorporating game design elements such as points, levels, and rewards into non-game contexts [1]. Studies have shown that gamified learning environments can increase participation and enhance learning outcomes, although their effectiveness depends on proper design and user context [2].

Simultaneously, mobile learning (M-learning) has transformed the accessibility of education by enabling learners to access content anytime and anywhere. The widespread adoption of smartphones has made mobile platforms a powerful medium for delivering interactive and personalized learning experiences [3]. Mobile-based systems are particularly suitable for short, engaging, and practice-oriented activities such as arithmetic training.

Another important advancement in educational technology is the development of adaptive learning systems. These systems personalize the learning experience by adjusting content difficulty and feedback based on individual user performance. Personalized gamification has been shown to significantly improve motivation, engagement, and cognitive outcomes compared to generic approaches [4].

In parallel, Vedic Mathematics offers an alternative approach to conventional arithmetic by providing efficient mental calculation techniques based on simple rules and patterns. These methods reduce computational complexity and improve calculation speed and accuracy, making them highly suitable for rapid problem-solving [5]. Despite its advantages, Vedic Mathematics has not been widely integrated into modern digital learning platforms.

Although existing research has explored gamification, mobile learning, and adaptive systems independently, there is limited work that combines all these approaches specifically for mental arithmetic training. Furthermore, most current applications lack integration with efficient computational techniques such as Vedic Mathematics, resulting in missed opportunities to enhance both engagement and performance.

To address these limitations, this project proposes a gamified, adaptive mobile application for learning Vedic Mathematics. The system incorporates interactive challenges, dynamic difficulty adjustment, and reward-based progression to create an engaging and effective learning experience. By combining gamification, personalization, and efficient mathematical techniques, the proposed solution aims to improve both user motivation and computational proficiency.

The main contributions of this work are as follows:

- Development of a gamified mobile application for arithmetic learning.

- Integration of Vedic Mathematics techniques for efficient computation.
- Implementation of adaptive difficulty based on user performance.
- Enhancement of user engagement through streaks, rewards, and interactive challenges.

2 Literature Review

2.1 Gamification in Education

Gamification has also become an effective strategy to increase user interaction and engagement. motivation outside of games. Deterding et al. referred to gamification as the use of non-game elements like points, badges and leaderboards in game design. In order to generate gameful experiences, we make some modifications to them. This initial work established the theoretical foundation of gamified systems but did not exist in practice. In this connection, Hamari et al. reviewed a wide range of empirical work. research on gamification and found that gamification tends to result in better user involvement, interest, and involvement. But the study that was highlighted is that the efficacy of gamification is situation specific and depends on user characteristics. Additionally, Ivankovi c et al created an adaptive and gamified learning system known as CodeyFor coding education. The system included such features as experience point (XP), streaks, leaderboards, and adaptive difficulty. However, their findings are evidence of increased student interactions and better learning results. They assessed that more large-scale validation was required.

Table 1 Comparison of Existing Literature

Author (Year)	Focus Area	Methodology	Key Findings	Limitations
Deterding et al. (2011)	Gamification Theory	Conceptual study	Defined gamification and game elements	No empirical validation
Hamari et al. (2014)	Gamification Review	Literature review	Gamification improves engagement	Context-dependent results
Ivanković et al. (2025)	Gamified Learning System	System development + testing	Improved engagement using XP, streaks	Limited sample size
Xiao & Hew (2024)	Personalized Gamification	Experimental study	Personalization improves motivation	Requires complex user modeling
Crompton (2013)	Mobile Learning	Historical analysis	Enables flexible learning	Lacks implementation focus
Patel & Parekh (2021)	M-Learning Adoption	Survey-based study	High mobile usage in education	Limited scalability discussion
Al-Fahad (2009)	User Perception	Survey analysis	Positive student perception	Device limitations
Träff et al. (2025)	Arithmetic Learning	Longitudinal study	Cognitive + number skills important	Focused on children only
Dowker (2019)	Cognitive Differences	Theoretical + empirical insights	Individual differences affect learning	Not app-based
Nawale et al. (2026)	Vedic Mathematics	Experimental analysis	Faster and accurate calculations	Limited digital integration

2.2 Personalized and Adaptive Learning Systems

Recent research has highlighted the importance of personalization in gamified learning environments. Xiao and Hew investigated personalized gamification and compared

it with traditional one-size-fits-all approaches [4]. Their study demonstrated that personalized gamification significantly improves motivational, behavioral, and cognitive outcomes, as it aligns game elements with individual learner preferences.

Adaptive learning systems further enhance this personalization by dynamically adjusting content and difficulty based on user performance. The Codey system also incorporated adaptive mechanisms such as feedback loops and difficulty scaling, enabling learners to progress at their own pace [6]. Despite these advancements, designing effective adaptive systems remains challenging due to the complexity of modeling user behavior.

2.3 Mobile Learning (M-Learning)

Mobile learning (M-learning) has transformed the educational landscape by enabling learning anytime and anywhere. Crompton provided a historical overview of mobile learning and highlighted its evolution toward learner-centered education [7]. The study emphasized the importance of accessibility and flexibility offered by mobile platforms.

Patel and Parekh examined the adoption of mobile learning in academic environments and reported that the widespread use of smartphones has significantly increased access to educational resources [8]. Their findings suggest that mobile learning systems must be optimized for usability and engagement to maximize effectiveness.

Additionally, Al-Fahad analyzed students' attitudes toward mobile learning and found that learners generally perceive mobile learning positively due to its convenience and ability to enhance communication [3]. However, challenges such as device limitations and potential distractions were also identified.

2.4 Mathematics Learning and Cognitive Factors

The cognitive aspects of learning mathematics need to be understood to develop effective learning tools. Träff et al. studied mental arithmetic skill development and concluded that both domain-specific number skills and domain-general cognitive abilities contribute to arithmetic skills. Their study also revealed that learning and performing require different cognitive processes.

Dowker studied individual differences in arithmetic abilities and emphasized how arithmetic performance varies significantly from person to person due to cognitive, emotional, and experiential factors.

2.5 Vedic Mathematics and Computational Efficiency

Vedic Mathematics offers an alternative approach to traditional arithmetic methods by using simplified techniques for faster computation. Nawale et al. demonstrated that Vedic Mathematics significantly improves computational speed and accuracy by reducing cognitive load and simplifying calculations [5]. The study also highlighted its applicability in modern domains such as engineering and computer science.

The integration of Vedic Mathematics into modern educational systems has the potential to enhance problem-solving skills and promote efficient mental calculation techniques. However, its adoption in digital learning platforms remains limited, indicating a gap in current research.

Table 2 Feature Comparison of Existing Approaches

System / Study	Gamification	Personalization	Mobile Support	Adaptive Difficulty	Math Focus
Deterding et al.	Yes	No	No	No	No
Hamari et al.	Yes	No	No	No	No
Codey System	Yes	Yes	No	Yes	No
Xiao & Hew	Yes	Yes	Yes	Yes	No
M-Learning Studies	No	No	Yes	No	No
Arithmetic Studies	No	Yes	No	No	Yes
Vedic Math Study	No	No	No	No	Yes
Proposed System	Yes	Yes	Yes	Yes	Yes

2.6 Summary and Research Gap

According to the literature, gamification, personalization, and mobile learning are highly effective in improving the outcome of education. Although gamification can be used for better engagement and motivation, personalization can be used for better results by considering the differences between individuals. Mobile learning can also be used for better accessibility, making it a suitable option for modern education.

However, existing systems are mostly based on general gamification techniques, and efficient mathematical techniques such as Vedic Mathematics are not considered in these systems. Moreover, research on adaptive gamification with mobile devices for mental arithmetic training is still in its infancy. This project proposes a gamified adaptive mobile application that uses Vedic Mathematics for better computational efficiency and engagement.

Table 3 Identified Research Gaps

Existing Limitation	Proposed Solution
One-size-fits-all gamification	Personalized gamification using user preferences
Lack of mobile-first systems	Develop mobile-based learning application
No integration of Vedic Mathematics	Incorporate Vedic techniques for fast calculation
Limited adaptive difficulty systems	Dynamic difficulty adjustment based on performance
Lack of engagement tracking	Use streaks, XP, and rewards system

3 Methodology

3.1 System Overview

The proposed system is a mobile-based Vedic Mathematics learning system, which can improve users' arithmetic skills. The system is based on a modular and service-oriented architecture, using an Android-based front-end and Firebase Authentication and Cloud Firestore for the backend.

The system is composed of a user interaction component, learning and challenge parts, common computational parts, and a cloud-based database for storage.

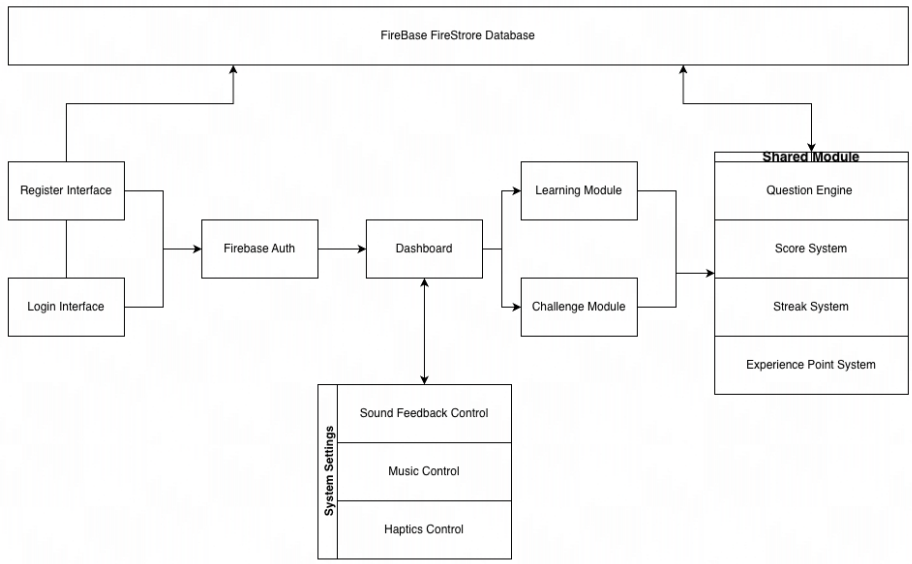

Fig. 1 System Architecture of the Proposed Application

3.2 System Architecture

The system architecture is composed of three main layers. The front-end client, which is based on Android, is responsible for user interaction, providing real-time feedback, and using a joystick for input. The second layer is composed of the functional layer, which contains the Practice Module and Challenge Module. The backend layer utilizes Firebase Authentication for user management and Firestore for storage.

3.3 Overall System Workflow

As depicted in Figure 2, the system starts from user authentication through the login or registration process. Once the user successfully authenticates the account, the system will redirect the user to the dashboard where the user can select between the Practice Module and the Challenge Module. Other features are also incorporated into the system.

3.4 Practice Module Workflow

The Practice Module is for conceptual understanding of Vedic Sutras. The workflow of this system starts from choosing a Sutra based on user progress. The system will demonstrate, and then users can solve step-by-step problems. The response of users

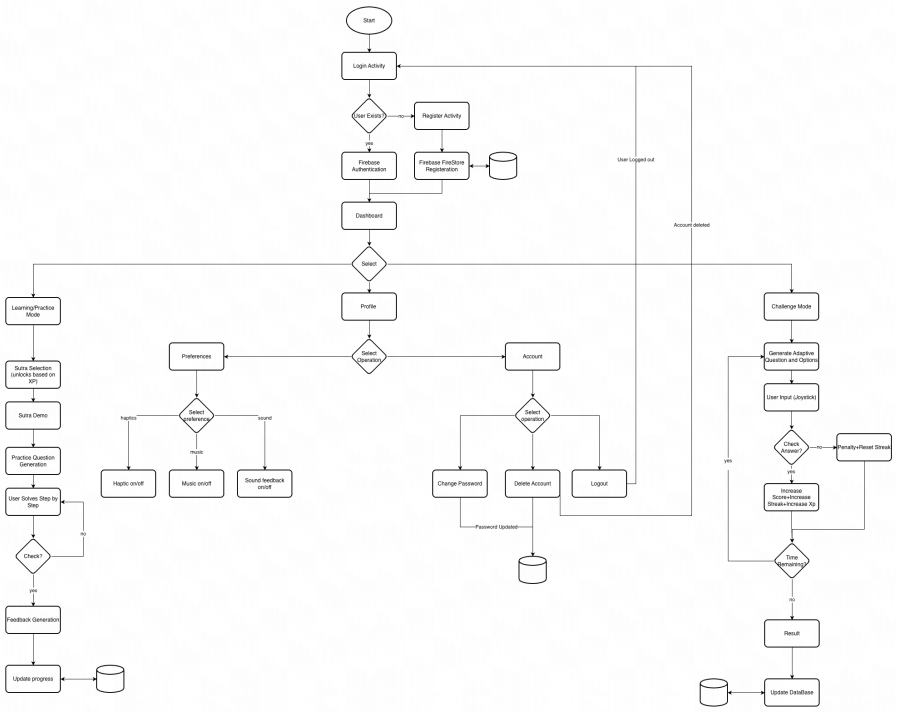

Fig. 2 Overall System Workflow of the Proposed Vedic Mathematics Application

will be checked, and feedback will be generated. If it is wrong, users can try again. After correct completion, it will be updated in the database.

3.5 Challenge Module Workflow

The Challenge Module uses gamification through time-bound problem solving. The questions are generated dynamically depending on the difficulty level. The user input is taken through a joystick. The accuracy of the input and the updating of the score, experience points, and streak are checked. The process continues until the time expires. Then, the results are computed and saved.

3.6 Adaptive Difficulty Mechanism

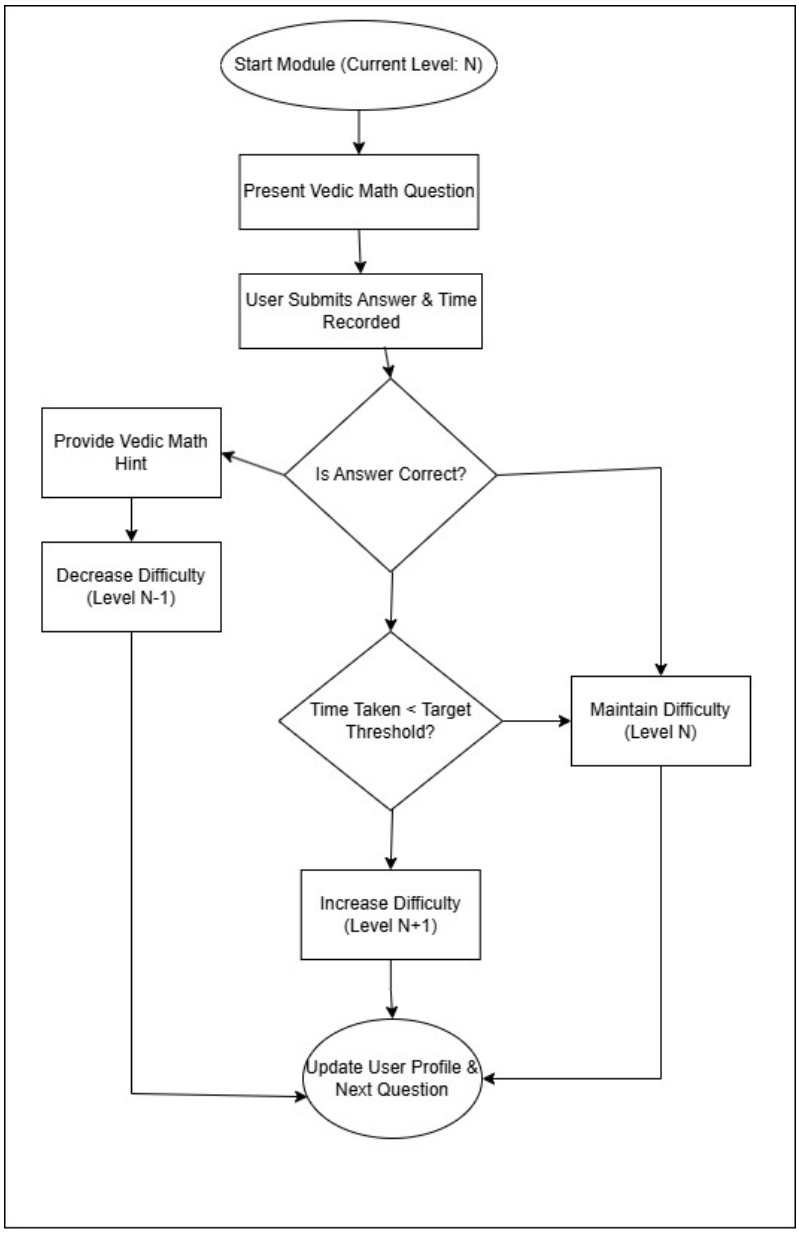

Fig. 3 Adaptive Difficulty Adjustment Mechanism

It also has an adaptive difficulty level that depends on the user's performance, as illustrated in Figure 3. The difficulty level increases if the user correctly answers the

question within the desired time, remains the same if the answer is correct but given at a slower rate, and decreases if the answer is incorrect.

3.7 Question Generation Strategy

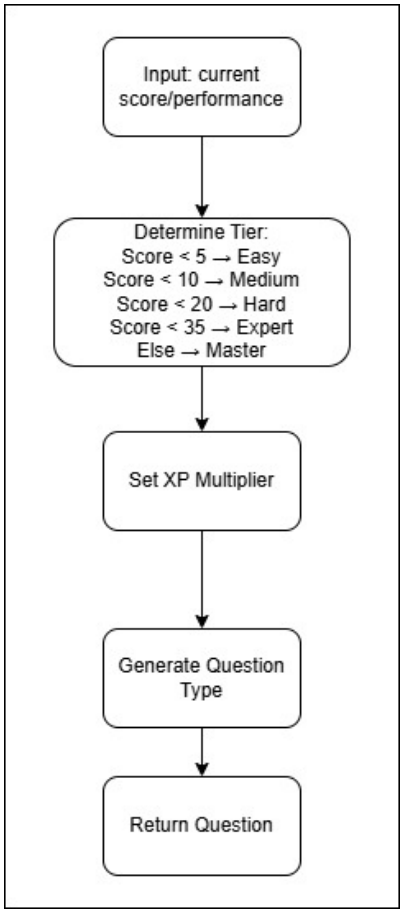

Fig. 4 Question Generation Process Based on Difficulty Levels

The question generation module identifies the difficulty tier depending on the user's score, as shown in Figure 4. Each difficulty tier has a corresponding XP multiplier and question type. Vedic arithmetic questions are generated according to this.

3.8 Joystick-Based Input Handling

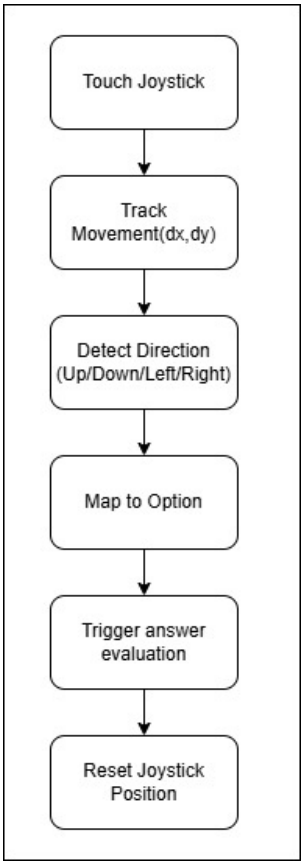

Fig. 5 Joystick Input Processing Mechanism

It (5) uses a joystick input mechanism to improve interactivity. The input mechanism works as follows. The user moves their touch input, and this is translated into a direction. The directions are used as answer choices. The answer choice is then evaluated, and the joystick returns to its original position.

3.9 XP and Streak Management

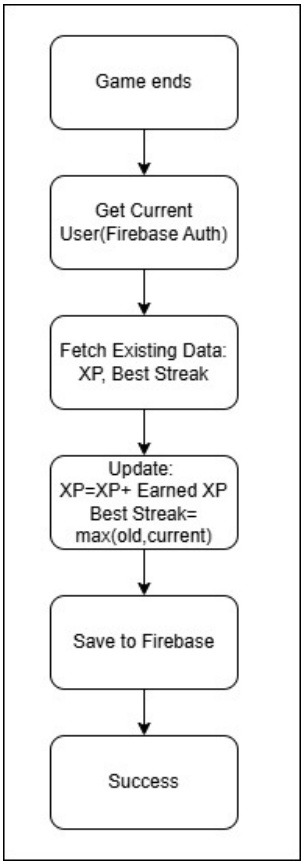

Fig. 6 XP and Best Streak Update Process

The gamification system keeps track of experience points (XP) and the best streak for the user. At the end of each session, it retrieves all the data from Firestore for the user, as shown in Figure 6. The experience points are incremented by the earned value, and the streak is updated as the maximum of the previous best streak and the new streak. The new values are stored.

3.10 Data Persistence and Cloud Integration

Firebase Firestore is used for real-time storage and synchronization of data. The database stores user credentials, XP, streaks, and performance information. Firebase Authentication is used for secure management of user accounts. This approach offers scalability, reliability, and simplicity in backend management.

3.11 System Integration

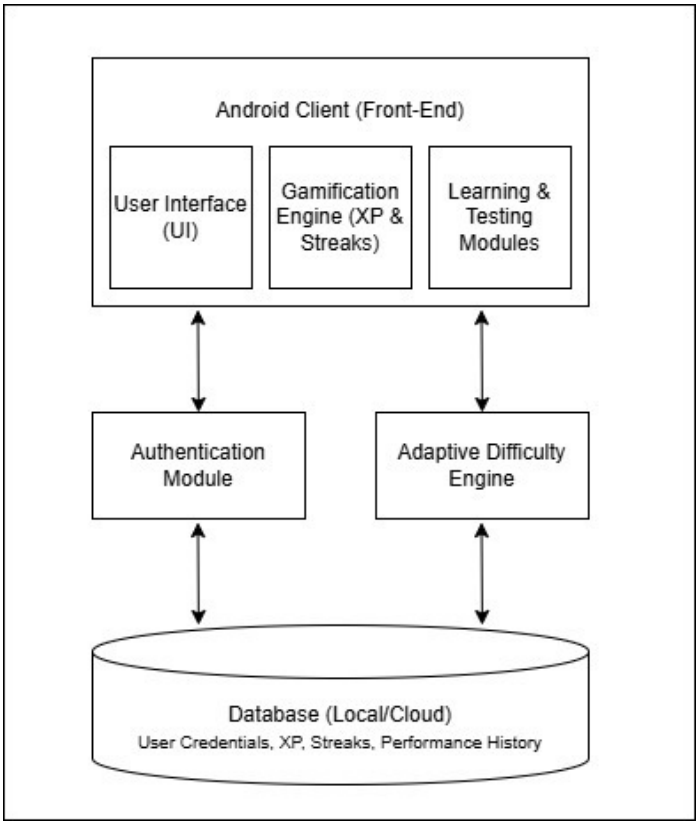

Fig. 7 Abstracted System Integration Model

As a whole, the system comprises different components, which are the user interface, adaptive difficulty, gamification, and database (firebase firestore), as shown in Figure 7.

Presentation Layer (Front-End): In this layer, the user interface was designed using Extensible Markup Language (XML) for faster response and better accessibility for different screen sizes of the mobile devices used by the users. The activities of the android are responsible for handling the main states of the application, which are login and dashboard. In addition, the application uses a fragment for content switching while a user is either learning or testing, as opposed to reloading the screen.

Application Logic Layer (Middleware): In this layer, the logic of the application was designed using Java, based on ViewModel and LiveData of the android operating system, for observing the change of data in real-time, ensuring that as soon as the user's scores or the number of wins in a row are updated in the database, they are reflected on the user interface immediately.

Data Management and Security: To manage user credentials and personalized data in a secure manner, the system utilizes Firebase Authentication for user verification. The passwords and credentials are managed securely by the Firebase Authentication mechanisms. To manage the data, the system utilizes the Cloud Firestore database for storing user-related data, including experience points (XP), streaks, and performance metrics. The data is stored in the cloud, allowing real-time synchronization of the data, thereby eliminating the need to store the data locally using storage mechanisms.

3.12 Integration of Vedic Mathematics Modules

The educational part of the app involves digitization of traditional Vedic math formulas and rules, such as Urdhva Tiryakbhyam (Vertically and Crosswise) and Nikhilam Navatashcaramam Dashatah (All from 9 and the last from 10), which simplify complex multiplication and squaring. Each learning module has two distinct phases:

Instructional Phase: Users are exposed to a new Vedic math rule with visual and stepwise depiction of the computation process. This reduces cognitive overload for learning new techniques.

Application Phase: Users get to engage in interactive and time-constrained activities to apply their learning of these techniques, thereby bridging the gap between learning and fast mental calculation.

3.13 Adaptive Difficulty Algorithm

In order to cater to the various learning needs and maintain the optimal state of engagement, a dynamic adaptive difficulty engine has been engineered. The user performance is assessed on two key factors: accuracy and response time.

The performance score for the user is computed by the engine after the user completes the question. In the case where the user is able to solve the problem correctly within a certain threshold (for example, 80% accuracy and an average response time of ≤ 5 seconds), the difficulty is increased by providing the user with more digits and/or applying advanced Vedic math techniques.

On the contrary, if the user's accuracy is less than a certain threshold, the difficulty is decreased by providing the user with certain hints and/or reviewing the relevant Vedic Sutra.

3.14 Gamification and Retention Engine

In an effort to compensate for the lack of motivation that is normally present in the traditional arithmetic drilling approach, the entire gamification system was incorporated.

Experience Points (XP) and Level Progression: The user is rewarded with various amounts of XP depending on the pace and accuracy of their response. By accumulating XP points, the user can move through various levels of the entire learning curriculum.

Daily Streak: A system was implemented to encourage the user to develop the habit of learning every day. By completing at least one mathematical problem, the user can increase the daily streak counter. A lack of activity for 24 hours resets the counter to zero.

Reward System: The user is rewarded by receiving badges conditionally upon the completion of certain milestones, such as "Master of Multiplication" and "7-Day Streak".

4 Software Requirement

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